

Suggested FYP topics for 2026-27

by

Dr. YIP Kit Sang Danny

Email: ksyip@ie.cuhk.edu.hk

Office: SHB 735

Project code :	KSY1
Project title :	AI-Powered Integrity Surveillance: Integrating Linguistic Obfuscation and Acoustic Prosody to Quantify Economic and Corporate Communications
No. of students :	1 group of 2 students
Description of project :	1. Forensic Linguistic Feature Extraction: Analyzing the complexity, readability, and sentiment of earnings call transcripts to detect signs of intentional clouding. 2. Acoustic Prosody and Behavioral Signal Processing: Extracting non-verbal cues from CEO speech patterns to identify psychological indicators of anxiety or avoidance during the Q&A sessions of performance reviews. 3. AI Integration: Developing a multimodal deep learning model that detects financial fraud among different data streams 4. Risk Quantification and Dashboard Implementation: Designing a decision-support system that aggregates these signals into a Dashboard. 5. This topic is reserved for a group of 2 students.
Industrial collaboration :	No.

Project code :	KSY2
Project title :	Personalized Risk Stratification and Management Systems for the Future of Geriatric Care
No. of students :	1 group of 2 students
Description of project :	1. Geriatric Digital Content and Data Acquisition: Systematically collecting and curating digital health resources and elderly-specific datasets. 2. Heterogeneous Multimodal Data Fusion: Integrating diverse data types (e.g., physiological, behavioral, environmental) from disparate sources into a unified framework. 3. AI-Powered Stratification and Management Engines: Developing machine learning models to automate risk assessment and personalized care interventions. 4. System Design and Application Implementation: Architecting and deploying a user-centric software application to interface with the risk management system. 5. This topic is reserved for a group of 2 students.
Industrial collaboration :	No.

Project code :	KSY3
Project title :	A Multi-Model Ensemble System for Predicting Match Outcomes and Betting Value Identification
No. of students :	1 group of 2 students
Description of project :	1. Data Acquisition and Feature Engineering: Identify and aggregate diverse data sources (historical results, player statistics, and real-time match data) to define key variables for football match analysis. 2. Market Analysis and Value Identification: Analyze various betting markets to calculate implied probabilities and identify values where market odds deviate from statistical predictions. 3. Model Development and Evaluation: Select, train, and benchmark multiple Machine Learning algorithms to determine the most accurate architecture for outcome forecasting. 4. System Deployment: Deploy the optimized model as a functional web or mobile application, providing users with real-time predictions and automated betting insights. 5. Predictive Factor Analysis: Conduct a feature importance study to identify the primary drivers of winning outcomes. 6. This topic is reserved for a group of 2 students.
Industrial collaboration :	No.

Project code :	KSY4
Project title :	Quantifying Resilience: A Comparative Analysis of Statistical and Data-Driven Approaches Against Data Poisoning Attacks
No. of students :	1 group of 2 students
Description of project :	1. Systematize poisoning attack vectors. 2. Implement dual-stream defense frameworks. 3. Establish comparative resilience metrics. 4. Develop a resilience decision matrix. 5. Deploy defense mechanism onto real applications.
Industrial collaboration :	No.

Project code :	KSY5
Project title :	The Path to Superhuman Agency: A Comparative Analysis of Reinforcement Learning Agents in Gaming
No. of students :	individual project, 1 group only
Description of project :	1. Market Analysis and Case Selection: Conduct a comprehensive survey of current gaming genres to identify and select optimal environments for algorithmic study. 2. Agent Architecture and Comparative Evaluation: Design and implement diverse autonomous agents to master selected game mechanics, followed by a rigorous benchmarking of their relative performance. 3. Optimization and Proficiency Analysis: Investigate the relationship between training convergence and agent efficacy, with a specific focus on the critical thresholds required to achieve superhuman performance. 4. This topic is reserved for one student.
Industrial collaboration :	No.

Project code :	KSY6
Project title :	Breakthroughs in Game Development
No. of students :	1 group of 2 students
Description of project :	1. Learn game programming. 2. Master game add-on, OOP, architectural patterns, design patterns, ... etc. 3. Learn state patterns on game control, e.g. lobby with multiple gamming maps. 4. Set up a game server. 5. Learn reinforcement learning for training game agents. 6. Apply recent technologies for game development 7. Record and edit videos on demonstrating your map. 8. Identify breakthroughs on your game product.
Industrial collaboration :	No.